Memory Is the Defence: Physics-Gated Continuum-Memory Receivers for Poisoning-Resilient Test-Time Learning

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***Abstract*—Neural receivers that learn at test time adapt to the channel from the pilots of every coherence block, so the pilot stream is a training set that an adversary can write to. This paper shows that such a receiver resists pilot poisoning when it consolidates into persistent memory only what an independent, label-free physical check verifies, and detects unverified blocks with the memory that check has protected. Citadel-Rx realises the principle in a DeepSIC receiver with a fast level fitted on every block and two slow levels updated every 8 and 32 blocks through a gated, clipped average of the faster states; three gates, each a ratio against a model-based reference on the same block, decide what is consolidated, and each CRC sub-block keeps the first of three detections that passes. Whatever the attacker does, the clip bounds the paired displacement of the slowest level by twice the clip per consolidation. A receiver that uses the same CRC test but keeps a single memory to roll back to fails, because one leaked update becomes its rollback state. With every setting fixed on calibration seeds and results on ten held-out seeds, white-box poisoning at 0 dB signal-to-interference ratio raises the symbol error rate of Citadel-Rx by [p1.citadel.wb0.excess?] over its own clean run, against [p1.online_deepsic.wb0.excess?] for tuned online DeepSIC; [p1.citadel.wb0.rejpct?]% of poisoned blocks are rejected at a [p1.citadel.wb0.frejpct?]% false-rejection rate, and without attack the error rate is [cmp.online_deepsic.clnrel?]% relative to tuned online DeepSIC. The attacks built against the defence, and the cases in which it does not help, are reported with the same protocol.**

***Index Terms*—Deep receivers, test-time learning, adversarial pilot poisoning, continual learning, physics-informed gating, MIMO detection, physical-layer security.**

I. Introduction

DEEP receivers replace or augment the model-based blocks of a MIMO detector with neural networks trained on data [1]–[4]. Since a network trained in simulation meets a different channel in the field, the strongest of them adapt at test time: online DeepSIC [5] re-trains its per-user networks on the pilots of every coherence block, and Meta-DeepSIC [6] meta-learns an initialisation from the pilots of past blocks so that the per-block fit converges faster. Their adaptation data are the only labelled symbols the receiver sees: a few hundred pilot bits per user per block.

An adversary within range of the base station can add a perturbation to the received pilot part of a block. Wu *et al.* [7] showed that a perturbation shaped through an unrolled copy of the receiver's adaptation step raises the post-adaptation symbol error rate (SER) of online DeepSIC by an order of magnitude, that a perturbation crafted on a surrogate still transfers, and that Meta-DeepSIC, which keeps the poisoned pilots in its support set, does not recover after the attack stops. The receiver cannot verify its adaptation data, because the pilots are the labels; what it can verify is physical consistency: whether its decisions fit a linear channel model, whether the CRC of the decoded data passes, and whether the estimated channel moved as a channel does. And the harm of poisoning lies less in the attacked block than in what the receiver keeps.

TABLE I
Positioning. Attack surface: what the adversary writes to.

Class	Adapts	Memory	Attack surface / defence
LMMSE	no	–	pilot estimate / –
Pilot-spoofing detection [8], [10], [11]	no	–	pilots / detection only
Online DeepSIC [5]	every block	last block	pilots / –
Meta-DeepSIC [6]	every block	pilot buffer	pilots, persistent / –
Code-verified online training [12], [13]	on decoded data	last block	labels / code check, no persistent gate
CRC-Rollback (Section VI-B)	every block	last accepted	pilots / CRC gate, rollback
Filtering and restoring TTA [14]–[17]	every batch	weights, anchor	batch / entropy filter, restore; i.i.d. batches
Robust TTA [22], [23]	every batch	batch statistics	batch / robust statistics
Anchored fine-tuning [19]–[21]	every step	prior weights	data / pull to a prior
Nested, continuum memory [26], [27]	every token	multi-rate	input / –
Citadel-Rx (this paper)	every block	three rates, gated	pilots / physics gates, protected fallback, bounded retention

This paper argues for one principle: *a test-time receiver resists pilot poisoning when it consolidates into persistent memory only what an independent, label-free physical check verifies, and detects unverified blocks with the memory that check has protected.* Both halves are necessary. The simplest receiver with the first half alone, online DeepSIC with a CRC test and rollback to its last accepted weights, fails: one poisoned update that passes the test becomes the state it rolls back to, and every later clean update is judged against it (Section VI-B). Citadel-Rx realises both halves with memory at three rates, a fast level that fits every block and two slow levels that change only through a gated, clipped average of the faster states; multi-rate memory is one realisation of the principle, and the ablation shows which of its parts are necessary.

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Code, configuration files, every run record, the statistics and the scripts that regenerate every table, figure and number of this paper are available at https://github.com/abraheemrashid99-sys/Citradel-Rx; one command re-runs the campaign and one rebuilds the paper (Section V).

Table I places the design among the work it draws on. Pilot spoofing is an established physical-layer threat [8], [9], with detectors that flag it [10], [11] but no learning receiver to protect. Decision-directed training on decoded data verified by the code is the closest precedent for using the code check to decide what a receiver learns from [12], [13]; it verifies labels, not what enters a persistent memory. Test-time adaptation in vision filters, anchors or restores what enters the adapted weights [14]–[18], anchors fine-tuning to a prior [19], [20] or averages a teacher [21], and has its own poisoning problem [22] with robust-statistics defences [23]; these methods assume i.i.d. batches and are compared below in receiver form. Byzantine-robust aggregation [24] needs many gradient vectors per step, and certified defences against data poisoning [25] bound the loss of a convex learner; here the learner is a network that sees one correlated block at a time. Nested learning [26] and Titans [27] keep memory at several rates but do not decide what enters the slow rates, and gated fast-to-slow consolidation echoes complementary learning systems [28], [29]. Adversarial examples against deep physical-layer models [30] and the methodology of attacks adapted to a defence [31], [33], [34] shape the evaluation. No published deep receiver combines multi-rate memory, physics-gated consolidation and a retention bound.

The contributions are as follows.

1) **The principle and a receiver that realises it** (Section III): gated, clipped consolidation into two slow levels, a fallback that detects a rejected block with the slow memory, and per-sub-block CRC selection among the fast, slow and LMMSE decisions; the ablation identifies the slow level, the fallback and the CRC gate as necessary.
2) **Retention bounds for the algorithm as implemented** (Section IV): a design bound, with no measured quantity, on the paired displacement of the slowest level under any attacker, and a structural bound for unequal gate verdicts computed from the paired trajectories. A first-version error in the consolidation rule, which broke the convexity the bounds rely on, is corrected.
3) **Relative physics gates with a derived CRC slack** (Section III-D): three label-free tests whose statistics are ratios against a model-based reference on the same block, and a CRC threshold derived from the pass-fraction statistics, which adapts to the block length.
4) **An evaluation built to be hard to pass** (Sections V–VI): tuned baselines, among them CRC-Rollback, anchored, decision-directed and EATA- and CoTTA-style receivers; attacks built against the defence (soft gates, deep unrolling, restarts, expectation over noise, joint pilot and data poisoning of the references, denial of adaptation, optimised slow drift, a rank-limited physical attacker, a grey-box surrogate and a query attacker); settings fixed on calibration seeds by rules stated in advance and results on held-out seeds; randomised attack windows; and the excess SER over a matched clean run with block-bootstrap intervals and seed-level tests.

Section II gives the system and threat model, Section III the receiver, Section IV the bounds, Section V the protocol, Section VI the results, Section VII their scope, and Section VIII concludes.

TABLE II
NOTATION.

U, N	users, base-station antennas
N_p, N_d, L, n	pilot and data symbols per user and block; symbols per CRC sub-block; sub-blocks per user and block
$\theta^{(k)}, K, C_k$	parameters of memory level k; number of levels; period of level k in blocks ($C_1 = 1$)
$\eta_b, \eta^{(k)}$	fast step size of block b; consolidation rate of level $k \geq 2$
a_b, λ, δ	anchor (slow projection); proximal weight; delta-rule decay
γ_b, τ_i, ω	gate verdict of block b; threshold of gate Gi; soft-step width
$F_{\mathrm{nn}}, F_{\mathrm{ref}}, p_{\mathrm{ref}}$	CRC pass fractions of the fast level and the reference; reference symbol-error probability
ρ, z, κ_k	relative CRC slack; statistical margin; consolidation clip of level k
$\boldsymbol{E}_b, \epsilon_b, q$	pilot perturbation; its energy budget; fraction of attacked blocks in the window
$W, n(W)$	window length in blocks; number of level-K consolidations in it
$d_b, D_k(b)$	paired displacement of the fast level and of level k (attacked minus clean run)

II. System Model and Threat Model

A. Uplink and Adaptive Receiver

U single-antenna users transmit QPSK symbols to a base station with N antennas. Time is divided into coherence blocks $b = 1, 2, \ldots$; within block b the channel $\boldsymbol{H}_b \in \mathbb{C}^{N\times U}$ is constant and the received vector at symbol slot n is

$$\boldsymbol{y}_n = f(\boldsymbol{H}_b \boldsymbol{s}_n) + \boldsymbol{w}_n, \qquad \boldsymbol{w}_n \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}), \tag{1}$$

with f the identity or the per-antenna power-amplifier nonlinearity $f(x) = \tanh(0.5x)$ [6]. Each block carries N_p pilot symbols per user, known to the receiver, and N_d data symbols framed in CRC-protected sub-blocks of L symbols per user. The main channel is the spatial exponential decay (SED) model of [5], $[\boldsymbol{H}_b]_{ij} = c_{j,b}\, e^{-|i-j|}$, with a per-user cosine fading gain $c_{j,b}$; Table V lists the other families. Table II collects the notation.

The receiver is DeepSIC [5]: Q soft-interference-cancellation iterations, each a per-user multilayer perceptron that maps the received vector and the other users' soft decisions to a posterior over the user's symbol. Its parameters are adapted on the pilots of every block, $\theta_b = \Phi(\theta_{b-1}; \boldsymbol{Y}_b)$, by T steps of a first-order optimiser on the pilot cross-entropy, and then detect the data. Online DeepSIC starts every block from θ_{b-1}; Meta-DeepSIC starts from an initialisation meta-learned by MAML [37] on a buffer of past pilot blocks, refreshed every five blocks [6].

B. Threat Model

The adversary adds $\boldsymbol{E}_b$ to the received pilot matrix $\boldsymbol{Y}_b \in \mathbb{C}^{N\times N_p}$ of an attacked block, with $\|\boldsymbol{E}_b\|_F \leq \epsilon_b$ set by a signal-to-interference ratio (SIR),

$$\epsilon_b = \|\boldsymbol{R}_b\|_F \; 10^{-\mathrm{SIR}/20}, \tag{2}$$

where $\boldsymbol{R}_b = \boldsymbol{Y}_b$ (received energy, noise included, as in [7]) or $\boldsymbol{R}_b = \boldsymbol{S}_b \boldsymbol{H}_b^{\top}$ (users' signal energy only); both are reported. At SIR 0 dB the attacker's received pilot energy equals that of all U users together (plus the noise, with the first budget). As an order of magnitude, with a common path-loss exponent

TABLE III
ATTACKER KNOWLEDGE.

White-box	victim's current weights, adaptation map, gate thresholds and running statistics; differentiates through them
Grey-box	the defence and its settings, not the weights: a Citadel-Rx surrogate trained on the pilots it observes
Transfer	an online DeepSIC surrogate trained on the pilots it observes
Query	no model: the victim's per-sub-block ACK/NACK feedback, searched with natural-evolution strategies [35], [36]

α, users at a common distance and equal transmit powers, the attacker must be $U^{1/\alpha}$ times closer to the base station than the users: about 1.9 times for $U = 4$ and the line-of-sight urban-micro exponent $\alpha = 2.1$ below the breakpoint distance [39]. The attacker is synchronised to the frame and knows the pilot sequence, which is standardised. The attackers are:

- *A1, availability* [7]: projected gradient ascent [38] on the post-adaptation data loss through an unrolled copy of the adaptation map.
- *A1 built against the defence*: deeper unrolling (10, 50 steps), more steps and random restarts, the soft-gate objective (the data loss of the fast level, which the receiver emits if it accepts the block, times a differentiable surrogate of the probability that its gates accept, plus a reward for acceptance; the surrogate uses the defence's exact thresholds and running statistics with the hard steps replaced by soft ones, as in BPDA [31]), and an expectation over noise draws (EOT [32]).
- *A4, reference degradation*: a fraction α of the energy on the pilots and $1 - \alpha$ on the data, optimised jointly so that the model-based references the gates compare against degrade with the fast level.
- *Denial of adaptation*: a fixed channel-estimate spoof $\boldsymbol{E}_b = \boldsymbol{S}_b\boldsymbol{\Delta}^\top$, meant to make the gates reject every block, combined with a real abrupt channel change.
- *A3*, optimised slow drift*: small perturbations over hundreds of blocks that pull the receiver toward a spoofed channel while keeping its blocks accepted.
- *Physical attacker*: $\boldsymbol{E}_b = \boldsymbol{G}\boldsymbol{X}_b$ with $M \in \{1, 2\}$ attacker antennas, a channel $\boldsymbol{G}$ from the same family, and an error on the attacker's estimate of $\boldsymbol{G}$.
- *Data jamming*: the same energy on the data part, a control that separates persistent poisoning from plain interference.

Table III states what each attacker knows.

C. Metrics

The SER is the fraction of symbol slots in which at least one user is wrong [5]–[7]; the CRC is checked per user and sub-block, and the CRC gate uses the per-user pass fractions. The primary metric is the *excess SER*: the SER of an attacked run minus that of the matched clean run (same configuration, seed and channel draws, attack disabled, same machine), averaged over the attack window. Two runs of a pair are identical up to the first attacked block, so the excess isolates the attack from the difficulty of the blocks it hits. The gate is characterised by the rejection rate of poisoned blocks and the false rejection rate of clean blocks, each with its count and a Wilson interval

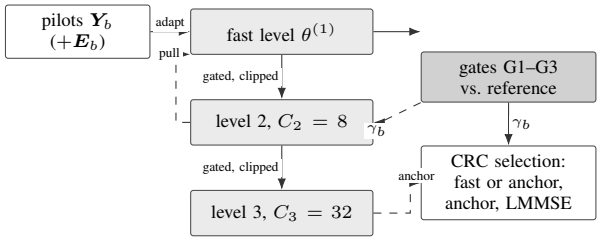


Fig. 1. Citadel-Rx. The fast level fits every block; the gates compare its decisions with a model-based reference on the same block; levels 2 and 3 receive gated, clipped averages of the faster states; a rejected block is reverted and detected with the anchor; every CRC sub-block keeps the first passing detection.

[45]. The paired displacement $D_k(b) = \left\|\theta^{(k)}_{\mathrm{adv},b} - \theta^{(k)}_{\mathrm{clean},b}\right\|$ measures what the attack left in level k.

III. CITADEL-RX

The receiver splits adaptation by time-scale: a fast level fits every block; slow levels change only through what the gates accept (Fig. 1, Algorithm 1).

A. Levels and the Fast Learner

The adaptable parameters $\theta \in \mathbb{R}^d$ (all DeepSIC weights, $d = 15\,628$ for 4×4) are kept at $K = 3$ levels with periods $C = (1, 8, 32)$ blocks. The fast level is adapted on every block by $T = 300$ steps of

$$\theta \leftarrow \theta - \eta_b \delta\, \theta\, \boldsymbol{\phi}\boldsymbol{\phi}^\top \quad \text{(delta-rule decay, per weight matrix)}, \tag{3}$$

$$\theta \leftarrow \theta - \eta_b \lambda(\theta - a_b) \quad \text{(proximal pull to the anchor)}, \tag{4}$$

$$\theta \leftarrow \mathrm{Adam}\big(\theta, \nabla_\theta \ell_b(\theta)\big), \tag{5}$$

with ℓ_b the pilot cross-entropy of the stage being adapted (stages are adapted sequentially [5]), $\boldsymbol{\phi}$ the normalised mean input of the stage, and $a_b = \frac{1}{2}\theta^{(2)}_{b-1} + \frac{1}{2}\theta^{(3)}_{b-1}$ the anchor. The step size follows the receiver's surprise $S_b = \|\nabla \ell_b(\theta_{b-1})\|^2$ relative to its moving average $\bar{S}_b$: $\eta_b = \eta_{\max}\, 2r_b/(1+r_b)$ with $r_b = \mathrm{clip}(S_b/\bar{S}_b, \frac{1}{8}, 8)$, so the rate is bounded by $2\eta_{\max}$.

B. Gated, Clipped Consolidation

At the end of every period of level $k{+}1$ the level receives the gated mean of the level-k states buffered since its last update, each weighted by the verdict $\gamma_j \in [0, 1]$ of its block (or the mean verdict of its period):

$$\theta^{(k+1)} \leftarrow \theta^{(k+1)} + \Pi_{\kappa_{k+1}}\Big(s\big(m - \theta^{(k+1)}\big)\Big),$$
$$m = \frac{\sum_j \gamma_j \theta^{(k)}_j}{\sum_j \gamma_j}, \qquad s = \eta^{(k+1)} \min\big(1, \textstyle\sum_j \gamma_j\big), \tag{6}$$
$$\Pi_\kappa(\boldsymbol{v}) = \boldsymbol{v} \min\{1, \kappa/\|\boldsymbol{v}\|\},$$

with $\eta^{(2)} = 1$, $\eta^{(3)} = \frac{1}{2}$, and no update when $\sum_j \gamma_j = 0$. The clip $\kappa = (8.5, 4.5)$ is the 99th percentile of the clean consolidation increments of levels 2 and 3 on calibration runs, so it is inactive on clean blocks and limits what any attacked period can move. The first version of the receiver divided the gated sum by $\max(1, \sum_j \gamma_j)$; when most blocks of a period

Algorithm 1 Citadel-Rx, one block b

Require: pilots $\boldsymbol{Y}_b$ (possibly poisoned), data $\boldsymbol{Y}_b^d$; levels $\theta^{(1..K)}$; buffers; running statistics
$a_b \leftarrow \sum_{k\geq 2} w_k \theta^{(k)}$; $\eta_b \leftarrow \eta_{\max} 2r_b/(1+r_b)$
reference: $\hat{\boldsymbol{H}}_b$ (LS, AR(1) tracker); $\hat{\boldsymbol{s}}_{\mathrm{mm}}$ (LMMSE); $\hat{\boldsymbol{s}}_a$ (anchor)
$\theta \leftarrow \theta^{(1)}$; T steps of (3)–(5) on $\boldsymbol{Y}_b$; $\hat{\boldsymbol{s}}_{\mathrm{nn}} \leftarrow$ detect$(\theta, \boldsymbol{Y}_b^d)$
$\gamma_b \leftarrow \min(\gamma_1, \gamma_2, \gamma_3)$: G1 residual ratio, G2 CRC ratio $F_{\mathrm{nn}}/F_{\mathrm{ref}}$, G3 innovation
if $\gamma_b < \frac{1}{2}$ **then**
    $\theta \leftarrow \theta^{(1)}$ (revert); $\hat{\boldsymbol{s}} \leftarrow \hat{\boldsymbol{s}}_a$ (fallback)
else
    $\hat{\boldsymbol{s}} \leftarrow \hat{\boldsymbol{s}}_{\mathrm{nn}}$; update the running statistics
end if
per user and sub-block: $\hat{\boldsymbol{s}} \leftarrow$ first CRC-passing of $(\hat{\boldsymbol{s}}, \hat{\boldsymbol{s}}_a, \hat{\boldsymbol{s}}_{\mathrm{mm}})$
$\theta^{(1)} \leftarrow \theta$; buffer $(\gamma_b, \theta^{(1)})$; for every k with $(b+1) \bmod C_{k+1} = 0$: (6)
return $\hat{\boldsymbol{s}}$

are rejected, which is the case under attack, this pulls the level toward the origin instead of leaving it nearly unchanged, and the update is then not the convex combination that the bounds of Section IV require. Every result of this revision uses (6).

C. Rejection, Fallback and CRC Selection

If $\gamma_b < \frac{1}{2}$ the block is rejected: the fast update is reverted, the reverted state enters the buffer with weight γ_b, and the data of the block are detected with the anchor, which the attacker has not reached. Each CRC sub-block of each user then keeps the first of three detections that passes its CRC: the gate's choice (fast level or anchor), the anchor, and LMMSE on the least-squares channel estimate. All three are computed for the gates anyway. Three optional states answer specific attacks and are evaluated with them: *abstain*, no consolidation when the reference itself has degraded (its pass fraction or residual is $z = 4$ standard deviations from its running level); *streak recovery*, after five consecutive rejections a re-fit from the anchor on CRC-verified data, kept if it verifies better than the reference; and a *drift guard*, a pull of level K toward the pre-trained weights by 5% per consolidation.

D. Relative Physics Gates

Every statistic is a ratio against a reference computed on the same block without labels: the LMMSE detector on the least-squares pilot estimate $\hat{\boldsymbol{H}}_b$ and the anchor detector, the better of the two. Ratios are normalised by their moving averages over accepted blocks and passed through a soft step of width $\omega = 0.1\tau$, tightened by up to 30% when $r_b > 1$.

G1, model residual. With $r(\hat{\boldsymbol{s}}) = \frac{1}{N_d}\sum_n \left\| \boldsymbol{y}_n - \hat{\boldsymbol{H}}_b \hat{\boldsymbol{s}}_n \right\|^2$, the statistic is $r(\hat{\boldsymbol{s}}_{\mathrm{nn}})/r(\hat{\boldsymbol{s}}_{\mathrm{ref}})$ against $\tau_1 = 1.25$.

G2, CRC pass rate. The pass fraction F_{nn} of the fast level over F_{ref} against the threshold τ_2 of the design rule below. The gate is tied to the availability attacker's own objective: an attack that raises the SER lowers the pass rate.

G3, channel prior. The innovation energy of the LS estimate against its AR(1) prediction, relative to its running level and floored by the $\chi^2_{0.999}$ quantile, against $\tau_3 = 4$.

Table IV gives the provenance of every setting.

TABLE IV
SETTINGS AND THEIR PROVENANCE. CALIBRATION: SEEDS 0–2 BY THE RULES OF SECTION V; FIRST VERSION: SET DURING THE DEVELOPMENT OF THE FIRST VERSION ON SEEDS 0–4. NO SETTING WAS CHOSEN ON THE TEST SEEDS 10–19.

Setting	Value; provenance
Q, MLP width, T, $\eta_{\max}$, Adam	3, 64·order, 300, $5 \cdot 10^{-3}$; DeepSIC reference [5]
K, C	3, $(1, 8, 32)$; first version; ablation in Section VI-G
$\eta^{(2)}, \eta^{(3)}$, λ, δ	1, $\frac{1}{2}$, 0.5, 0.1; first version; sensitivity in Section VI-I
τ_1, τ_3, ω, tightening	1.25, 4, 0.1τ, 30%; first version; sensitivity
CRC slack	relative $\rho = 1$ with floor $2 \cdot 10^{-3}$, or derived $z = 5$; mode chosen by calibration ([cal.chosen?])
κ	$(8.5, 4.5)$; 99th percentile of clean consolidation increments, calibration runs
Consolidation rule, CRC selection	(6); correctness fix; CRC selection kept by calibration
Baseline epochs, learning rate, weight decay	tuned on calibration seeds (Section V)
Attacker	PGD 20 steps of 0.25ϵ, 2 unrolled steps [7]; stronger variants in Section VI-C

IV. RETENTION BOUNDS

The bounds are stated for the algorithm as implemented, for a pair of runs on the same channel, noise and seed, with and without an attack that starts at block b_0 (the runs are identical before b_0). Let $n(W)$ be the number of level-K consolidations in $[b_0, b_0 + W)$, so $n(W) \leq \lceil W/C_K \rceil$; quantities of the clean run are primed.

Lemma 1 (convex consolidation)**.** *Without the clip,* (6) *writes the new level-$(k{+}1)$ state as the convex combination $(1-s)\theta^{(k+1)} + s\,m$ of its old state and the buffered states, with $s \leq \eta^{(k+1)}$ and m in their convex hull. With the clip the step is $\tilde{s} = s \min\{1, \kappa_{k+1}/\left\|s(m - \theta^{(k+1)})\right\|\} \leq s$, and the statement holds with $\tilde{s}$.*

Theorem 1 (retention)**.** *(i)* Design bound. *With the clip, for any perturbations and any gate verdicts,*

$$D_K(b_0 + W) \leq 2\kappa_K\, n(W). \tag{7}$$

(ii) Structural bound. *Let $\Delta(W)$ be the largest paired displacement of the buffered level-$(K{-}1)$ states in the window and, at the i-th consolidation, $\bar{w}_i, \bar{w}'_i$ the normalised weights, $\tilde{s}_i, \tilde{s}'_i$ the steps taken, m'_i the clean gated mean and r'_i the radius of the clean buffer around it. Then*

$$D_K(b_0 + W) \leq \big(\Delta(W) + \Omega(W)\big)\big(1 - (1-\eta^{(K)})^{n(W)}\big) + \sum_{i \leq n(W)} |\tilde{s}_i - \tilde{s}'_i| \left\| \theta_i'^{(K)} - m'_i \right\|, \tag{8}$$

with $\Omega(W) = \max_i \|\bar{w}_i - \bar{w}'_i\|_1\, r'_i$. With equal verdicts $\Omega = 0$, the sum vanishes, and $\Delta(W)$ is at most the largest paired fast-level displacement.

Proof. (i) Level K changes only at consolidations, and in each run by at most κ_K; the drift guard, when used, multiplies the paired difference by 0.95. (ii) At consolidation i, $\theta^{(K)}_{\mathrm{new}} - \theta'^{(K)}_{\mathrm{new}} = (1-\tilde{s}_i)(\theta^{(K)} - \theta'^{(K)}) + \tilde{s}_i(m_i - m'_i) + (\tilde{s}_i - \tilde{s}'_i)(m'_i - \theta'^{(K)})$. Since $\sum_j \bar{w}_{ij} = \sum_j \bar{w}'_{ij} = 1$, $m_i - m'_i = \sum_j \bar{w}_{ij}(\theta_j - \theta'_j) + \sum_j (\bar{w}_{ij} - \bar{w}'_{ij})(\theta'_j - m'_i)$, whose norm is at most $\Delta + \Omega$. Iterating $D \leftarrow (1-\tilde{s}_i)D + \tilde{s}_i(\Delta + \Omega) + e_i$ from $D = 0$ gives $D \leq (\Delta + \Omega)(1 - \prod_i (1 - \tilde{s}_i)) + \sum_i e_i$, and $\tilde{s}_i \leq \eta^{(K)}$. □

Bound (7) needs no measured quantity and holds for every attacker of this paper, including those that pass the gates; its

price is that a legitimate change of the channel moves the slowest level by at most κ_K per period. Bound (8) is evaluated on the paired trajectories in Section VI-E; it is tighter when the verdicts agree, and it is the form in which the gate enters: rejected blocks are reverted and enter with weight γ_b.

Remark 1 (contraction). If the block map satisfied $\|\Phi(\theta; \boldsymbol{Y}+\boldsymbol{E}, a) - \Phi(\theta; \boldsymbol{Y}, a)\| \leq G_\Phi \|\boldsymbol{E}\|$ and were jointly L-Lipschitz in (θ, a) with $L < 1$, the largest fast displacement after n_W accepted attacked blocks would satisfy $\Delta_f \leq G_\Phi \epsilon (1 - L^{n_W})/(1 - L)$. The sufficient condition is $L = L_\theta + L_a < 1$. Finite-difference probes of the fast learner, which the revision does not change, give $L_\theta + L_a$ between 1.08 and 1.39 for proximal weights λ from 0.5 to 4; probes can only underestimate a Lipschitz constant, so the condition fails and this branch cannot be used for the default receiver. The regret bound of the first version, which relied on it, is withdrawn; a last-layer-adaptable variant, for which the loss is convex, is evaluated in Section VI-H without a regret claim.

Design rule for the CRC gate. With i.i.d. symbol errors of probability p in a sub-block of L symbols and an r-bit CRC whose syndrome is uniform on error, $P(\text{pass} \mid p) = (1-p)^L + 2^{-r}(1-(1-p)^L)$. Inverting at F_{ref} gives p_{ref}, and the rule "reject if $F_{\text{nn}}/F_{\text{ref}} < \tau_2$" with $\tau_2 = \big((1 - p_{\text{ref}} - \delta)/(1-p_{\text{ref}})\big)^L$ tolerates an excess error probability δ over the reference. The relative slack sets $\delta = \max\{2 \cdot 10^{-3}, \rho\, p_{\text{ref}}\}$; the derived slack removes the floor and takes $\tau_2 = \min\{1 - z\sigma, \tau_2(\rho p_{\text{ref}})\}$, with σ the standard deviation of the normalised pass ratio on accepted blocks, floored by its binomial value $\sqrt{F(1-F)/n}/F$, so the threshold follows the block length. For n independent sub-blocks, $P(|\hat{F} - F| \geq t) \leq 2e^{-2nt^2}$ [43]. The formula is standard; its use here is to replace a tuned threshold by a target excess error.

Model mismatch and G1. Writing $\boldsymbol{y}_n = \hat{\boldsymbol{H}}\boldsymbol{s}_n + \boldsymbol{\nu}_n + \boldsymbol{m}_n$ with $\boldsymbol{m}_n$ the mismatch, the residual of a decision with error $\boldsymbol{e}_n$ is $\text{mean}\left\|\hat{\boldsymbol{H}}\boldsymbol{e}_n\right\|^2 + \text{mean}\,\|\boldsymbol{\nu}_n + \boldsymbol{m}_n\|^2$ plus a cross term that vanishes if decision errors are uncorrelated with noise and mismatch. This is an assumption, not a fact: errors occur when the noise is large. On clean blocks of the tanh channel the normalised G1 statistic has mean 1.000, 99th percentile 1.050 and maximum 1.080 over 285 blocks, and never reaches $\tau_1 = 1.25$.

V. Experimental Protocol

Receivers. LMMSE on the least-squares pilot estimate, and a Huber-robust version [42]; online DeepSIC [5]; Meta-DeepSIC [6]; *CRC-Rollback*, online DeepSIC with the CRC gate and rollback to the last accepted weights, and the same with all three gates; online DeepSIC with the CRC selection of Citadel-Rx, which separates the selection gain from the memory; L2-SP [19], EWC with the Fisher information of the pre-training data [20] and an EMA teacher [21]; CRC-verified decision-directed training, which adds the CRC-passing data symbols to the pilots [12], [13]; an EATA-style receiver, entropy filter and Fisher anchor [14], and a CoTTA-style receiver, teacher and stochastic restore [15]; a drift-detection receiver; a median-of-means gradient receiver after [23]; and a single-memory receiver with the fast learner of Citadel-Rx and no slow levels [27]. All DeepSIC receivers share the architecture and the pre-training.

TABLE V
Simulation setup.

System	$U \times N = 4 \times 4$, QPSK, SNR 10 dB; 4×16; 16-QAM at 20 dB
Block	$N_p = 100$ pilot and $N_d = 25\,000$ data symbols per user; $N_d \in \{1000, 5000\}$, $N_p \in \{32, 100\}$ in Section VI-H
CRC	CRC-8 of [40], $D^8 + D^7 + D^4 + D^3 + D + 1$; $L = 32$; $n = 781$ sub-blocks per user and block
Channel	SED with per-user cosine fading (periods 51, 39, 33, 21 blocks; depth 0.8) [5]; COST2100 traces of [6], [41] in calibration
Run	100 blocks (400 for A3*); 40-block attack window starting at a block drawn uniformly from 10–50 per seed
Seeds	calibration 0–2; test 10–19 (headline cell), 10–12 (other headline cells), 10–11 (other experiments), 10 (sensitivity)
Attacker	PGD 20 steps of 0.25ϵ through 2 unrolled steps [7], [38], unless stated

Calibration and tuning, fixed in advance. Online DeepSIC and Meta-DeepSIC are tuned over 300, 100 and 30 adaptation epochs, learning rates 10^{-3} and $5 \cdot 10^{-3}$ and weight decay 0 and 10^{-4} on clean runs of the calibration seeds; the lowest mean clean SER wins (online DeepSIC: [tune.onlinedeepsic.epochs?] epochs, learning rate [tune.onlinedeepsic.lr?], weight decay [tune.onlinedeepsic.wd?]; Meta-DeepSIC: [tune.metadeepsic.epochs?], [tune.metadeepsic.lr?], [tune.metadeepsic.wd?]), and every receiver built on online DeepSIC uses its setting. Citadel-Rx has three candidates, relative or derived CRC slack and, with the relative slack, a fast level restarted from the anchor every block; the rule, fixed before the runs, is the lowest mean of the clean SER, the A1 window SER at SIR 0 dB, and the A1 window SER on COST2100, among candidates with at most 2% clean false rejections (Table VI). The chosen settings are written to a file by the rule and read by every test run.

Statistics. Within a run, consecutive blocks are correlated (decorrelation lengths of [p1.citadel.wb0.decor?] blocks for the excess SER at the headline cell), so intervals are 95% moving-block bootstrap intervals [44] that resample seeds and then runs of twice the decorrelation length within each seed. Tests are at the seed level: an exact two-sided Wilcoxon signed-rank test on the per-seed means [46], whose smallest attainable p with ten seeds is 0.002, Holm-corrected over the four baselines of a column [47]. Rates carry their counts and Wilson intervals.

Reproducibility. The campaign ran on an Apple M4 laptop and a 4-core cloud machine, one single-threaded run per core, with disjoint sets of runs. Every record stores its configuration, seed, code version and timings; an attacked run and its clean twin always ran on the same machine. Pre-training is cached by configuration, seed, code and random-number state, which is exact (the repository's tests compare cached and uncached runs). One command re-runs the campaign and one recomputes every number and table of this paper.

TABLE VI
CALIBRATION OF CITADEL-RX ON SEEDS 0–2 (SER). SCORE: MEAN OF THE CLEAN, A1 AND COST2100 A1 COLUMNS.

Candidate	Clean	A1 window	COST2100 A1	False rej.	Score
(runs pending)					

VI. RESULTS

A. Headline Comparison

Table VII contains the main result. [note.p1?]

B. Why Rollback Is Not Enough

[note.c1?]

C. Attacks Built Against the Defence

[note.adaptive?]

D. Attacks on the References and on Adaptation

[note.refdeg?]

E. Retention Under Optimised Slow Drift

[note.retention?]

F. Other Baselines

[note.baselines?]

G. Ablations

[note.ablations?]

H. Block Length, Arrays, Modulation and Cost

[note.scen?]

I. Sensitivity

[note.sens?]

VII. DISCUSSION

[note.discussion?]

VIII. CONCLUSION

[note.conclusion?]

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The COST2100 traces are those released with [6].

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TABLE VII
EXCESS SER (ATTACKED MINUS MATCHED CLEAN RUN, ATTACK WINDOW) AT THE HEADLINE CELLS, TEST SEEDS; 95% MOVING-BLOCK INTERVAL AT WHITE-BOX SIR 0 DB (10 SEEDS; OTHER COLUMNS 3 SEEDS). CLEAN: SER OF THE CLEAN TWINS OVER BLOCKS 5–99. REJECTION AND FALSE REJECTION OF THE CITADEL-RX GATES AT WHITE-BOX SIR 0 DB. SCRIPT: REVISION_NUMBERS.PY.

Receiver	Clean	WB −10 dB	WB 0 dB	WB +10 dB	Transfer 0 dB	Jamming 0 dB	Rej.	False rej.
(runs pending)								

TABLE VIII
PAIRED SEED-LEVEL COMPARISON, CITADEL-RX MINUS BASELINE: ATTACK-WINDOW SER AT WHITE-BOX SIR 0 DB AND CLEAN SER (RELATIVE CHANGE), EXACT WILCOXON p, HOLM-CORRECTED OVER THE FOUR BASELINES.

Baseline	Seeds	Δ window SER	p	Δ clean SER	p
(runs pending)					

TABLE IX
ATTACKS BUILT AGAINST THE DEFENCE, SIR 0 DB UNLESS STATED, TEST SEEDS 10–11. [a] SOFT GATES, UNROLL 10, PGD 50, 2 RESTARTS, EOT OVER 2 NOISE DRAWS; [b] THE SAME WITHOUT SOFT GATES, WHICH ONLINE DEEPSIC DOES NOT HAVE.

Attack	Receiver	Seeds	Excess SER [95% CI]	Window SER	Rej.
(runs pending)					

TABLE X
JOINT PILOT AND DATA ATTACK ON THE REFERENCES (A4), WHITE-BOX SIR 0 DB, 6 PGD STEPS OVER ALL DATA SUB-BLOCKS, TEST SEEDS 10–11, WITH AND WITHOUT THE ABSTAIN STATE; JAMMING CONTROLS WITH THE DATA POWER OF A4.

Attack	State	Seeds	Excess SER [95% CI]	Window SER	Rej.
(runs pending)					

TABLE XI
ABRUPT CHANNEL REDRAW AT BLOCK 50, WITHOUT AND WITH THE DENIAL-OF-ADAPTATION SPOOF ON BLOCKS 40–89 (SIR 10 DB), AND A 6 DB SNR DROP AT BLOCK 50; TEST SEEDS 10–11. SER AFTER THE CHANGE (NO ATTACK), FALSE REJECTION AFTER IT, EXCESS SER OF THE SPOOF IN AND AFTER ITS WINDOW, SER AFTER THE SNR DROP.

Receiver	Seeds	SER after redraw	False rej.	Spoof excess	After spoof	SER after drop
(runs pending)						

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TABLE XII
RETENTION OF THE SLOWEST LEVEL: PAIRED DISPLACEMENT D_K (LARGEST IN THE WINDOW, MEAN OVER SEEDS) AGAINST THE STRUCTURAL BOUND (8) AND THE DESIGN BOUND (7). A1: PAIRED RUNS WITH LEVEL SNAPSHOTS, FIXED WINDOW 30–69, SEEDS 10–12. A3*: BLOCKS 30–329 OF 400, ENERGY RELATIVE TO THE PILOTS, SEEDS 10–11.

Attack	Seeds	Excess SER [95% CI]	Rej.	D_K	Struct.	Design
(runs pending)						

TABLE XIII
FURTHER BASELINES UNDER WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–11.

Receiver	Seeds	Clean SER	Excess SER [95% CI]	Window SER
(runs pending)				

TABLE XIV
ABLATIONS UNDER WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–11.

Configuration	Seeds	Clean SER	Excess SER [95% CI]	Rej.	False rej.
(runs pending)					

TABLE XV
BLOCK LENGTH, A 4×16 ARRAY, 16-QAM AND TWO CHEAPER VARIANTS, WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–11; TIME PER BLOCK ON ONE CORE OF THE LAPTOP.

Setting	Receiver	Seeds	Clean	Excess SER [95% CI]	Rej.	s/block
(runs pending)						

TABLE XVI
ONE-AT-A-TIME SENSITIVITY, WHITE-BOX A1 AT SIR 0 DB, TEST SEED 10: CLEAN SER, EXCESS SER AND CLEAN FALSE REJECTION WITH EACH SETTING HALVED AND DOUBLED.

	$\times 0.5$			$\times 2$		
Setting	Clean	Excess	False rej.	Clean	Excess	False rej.
(runs pending)						